

TRIGA Mark I: coupled neutronics and thermal-hydraulic reference calculation

An independent Cardinal / OpenMC / MOOSE / NekRS calculation of a TRIGA Mark I unit cell and bare core, produced without reference to the parallel GeN-Foam / OFFBEAT calculation, then compared against it.

ITEM	VALUE
Codes	Cardinal snapshot-20-10-27 · OpenMC 0.16.1-dev14 · MOOSE dddb15e48 · NekRS
Platform	Ubuntu container, 32 GiB, NVIDIA GTX 1080 Ti
Nuclear data	ENDF/B-VIII.0 (neutronics baselines) · ENDF/B-VII.1 (coupled production)
Independence	Cardinal results frozen at git tag pcm-independent-v1 before GeN-Foam was read
Repository	github.com/kavyawadhwa134/pcm-triga-cardinal
Status	Neutronics and conduction converged · CFD flow converged, thermal transient

SCOPE OF THIS DOCUMENT

This report presents verification and a cross-code comparison. **It is not a validation report.** No experimental data was used at any point, and agreement between two codes is not a substitute for measurement. Section 8 states the status of each verification level explicitly.

1 Summary

Cardinal has produced an independent, reproducible result set for the specified TRIGA Mark I problem. Monte Carlo neutronics is coupled to MOOSE heat conduction through Cardinal and converged over ten Picard iterations with exact power conservation. A standalone NekRS large-eddy simulation of the coolant subchannel reached a converged flow field but not a converged temperature field.

QUANTITY	VALUE	UNCERTAINTY	MODEL
k-infinity	1.384207	± 21 pcm	2-D unit cell, reflective
k-effective	1.102059	± 26 pcm	3-D bare core, vacuum
Leakage fraction	0.20149	± 0.00012	bare core
Peak fuel temperature	359.0 K	—	average pin, coupled
Peak clad temperature	336.2 K	—	average pin, coupled
Fuel temperature coefficient	-6.34 pcm/K	—	296–1000 K, computed
Pin power peak/average	2.2544	—	120 pins

PRINCIPAL FINDINGS

1 • Where the two calculations are methodologically comparable, they agree. The Monte Carlo pin cell differs from GeN-Foam by 280 pcm — inside the 500 pcm threshold set for this exercise.

2 • The k-effective difference is method, not error. It decomposes into +3920 pcm of homogenisation and +8320 pcm of diffusion-versus-Monte-Carlo on geometrically identical bare cores. GeN-Foam's own data independently measures a -4114 pcm method difference on the pin cell.

3 • The fuel temperature coefficient is the most consequential open item. Cardinal computes -7.76 pcm/K in GeN-Foam's operating band; GeN-Foam uses a constant -10.000 pcm/K whose origin is not documented in the committed files. This governs the transient result.

4 • Cardinal's CFD thermal field is not converged and its coolant temperatures are lower bounds, not results.

2 Scope, objectives and methodology

2.1 Objective

Produce an independently reproducible TRIGA Mark I reference calculation using the Cardinal ecosystem, to serve as a cross-code reference against a parallel calculation (GeN-Foam) performed in GeN-Foam and OFFBEAT. The two calculations were produced without reference to one another; only the scientific problem definition is shared.

2.2 Scope

In scope: steady-state eigenvalue neutronics for a 2-D unit cell and a 3-D bare core; coupled neutronics–conduction for a single pin; standalone coolant CFD.

Out of scope and not attempted: the +0.5 \$ reactivity transient (the project workflow specifies a steady-state production run and defines no transient phase); burnup; control-rod worth; experimental comparison.

Model reduction. The pin-level coupled model is a unit cell, not a full core. This was forced by the original 8 GiB development machine and retained after the move to a 32 GiB host so the coupled result remains comparable to the frozen baseline. All coupled temperatures therefore describe an **average pin**; Cardinal's own peaking factor of 2.2544 implies a hot channel is materially hotter.

2.3 Methodology

STAGE	CODE	METHOD
Neutronics	OpenMC	Monte Carlo, continuous energy
Solid conduction	MOOSE	Finite element, 6400 elements, stitched fuel–clad contact
Coolant	NekRS	Spectral element, implicit LES (HPFRT), polynomial order 7
Coupling	Cardinal	Picard iteration; power to MOOSE, temperature back to OpenMC

Thermal scattering $S(\alpha, \beta)$ for $c_H_in_ZrH$ and $c_Zr_in_ZrH$ is applied throughout. Without it the bound-hydrogen physics that defines a TRIGA is absent; §6 demonstrates it is present.

2.4 Specification

PARAMETER	VALUE	BASIS
Fuel outer radius	1.815 cm	specification
Clad outer radius	1.866 cm	specification
Lattice pitch (square)	4.235 cm	specification
Active height	60.0 cm	specification
Fuel U-ZrH1.65	6.0 g/cm ³	U-235 1.7, U-238 6.8, Zr 89.862, H-1 1.638 wt%
Clad SS304	8.00 g/cm ³	Cardinal assumption
Coolant	1.0 g/cm ³	H ₂ O, no boron
Thermal power	250 kW	specification
Power per pin (120 pins)	2083.33 W	Cardinal derivation
Coolant inlet	293 K, 0.2 m/s	specification

There is no fuel–clad gap: the specification gives two radii, so clad inner radius equals fuel outer radius. The MOOSE mesh stitches the two regions into direct thermal contact rather than separating them with a gap conductance.

3 Thermal results

Solid conduction in fuel and clad, with fission power supplied by OpenMC each Picard iteration and a convective boundary condition at the clad surface.

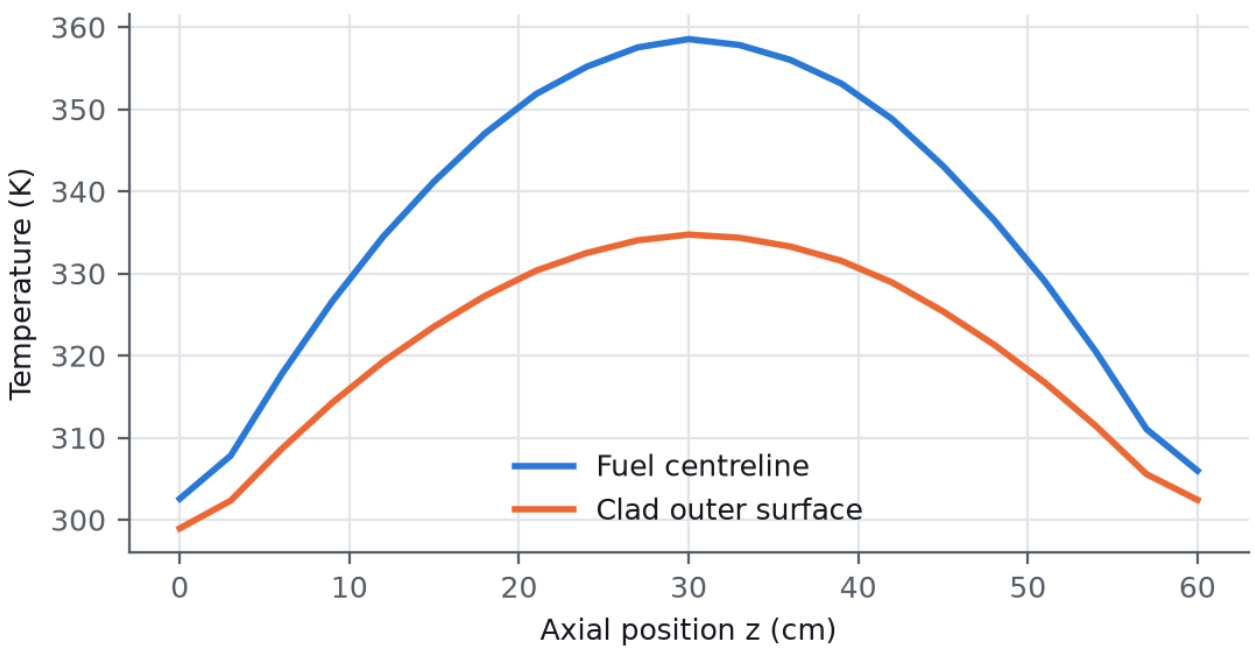


Figure 1 — Axial temperature profile. The profile is slightly top-skewed (305.98 K at the outlet against 302.56 K at the inlet) because the coolant heats as it flows upward.

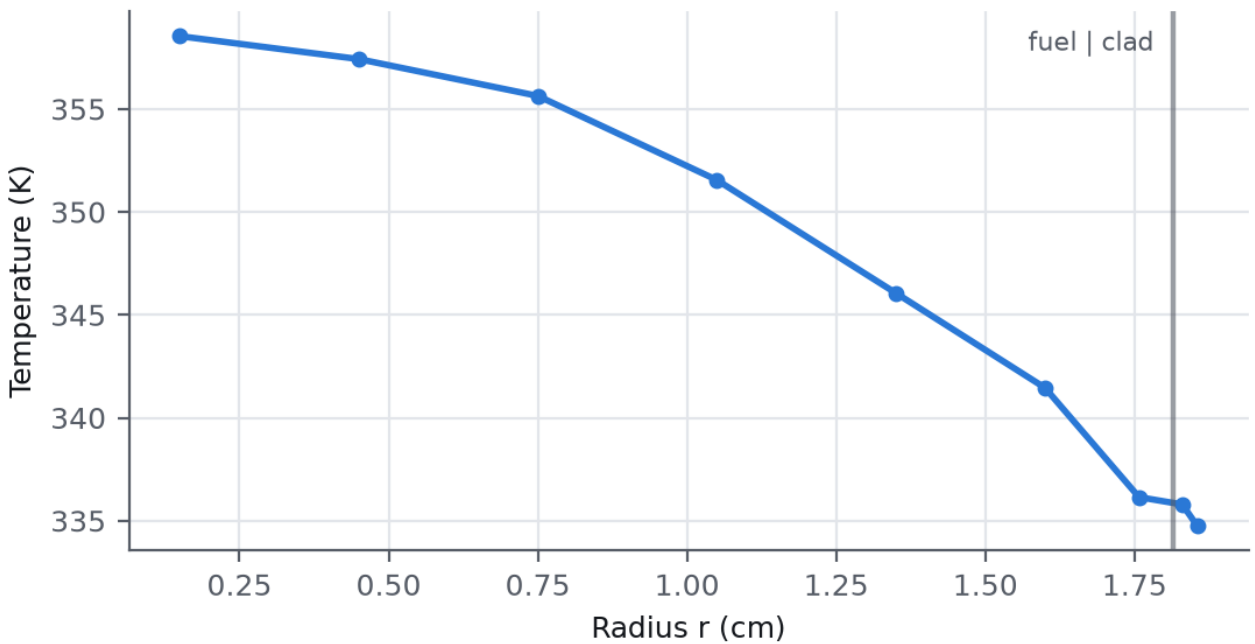


Figure 2 — Radial profile at mid-plane. Temperature is continuous across the fuel–clad interface at $r = 1.815$ cm, confirming the stitched contact behaves as specified.

3.1 Temperature drop budget

CONTRIBUTION	COMPUTED	ANALYTIC	SHARE	GOVERNED BY
Film (clad → coolant)	40.3 K	42.4 K	63%	correlation — weakest term
Fuel conduction	22.4 K	24.1 K	35%	$k_{\text{fuel}} = 18 \text{ W/m}\cdot\text{K}$

CONTRIBUTION	COMPUTED	ANALYTIC	SHARE	GOVERNED BY
Clad conduction	1.07 K	1.5 K	2%	$k_{\text{clad}} = 16 \text{ W/m}\cdot\text{K}$

The film drop dominates, and it is the term Cardinal knows least well — a Dittus–Boelter correlation evaluated at $Re = 4755$, which is transitional and below the range where that correlation is reliable. Replacing it with a converged NekRS solution is the single highest-value remaining improvement to the thermal result.

4 Numerical and physics verification

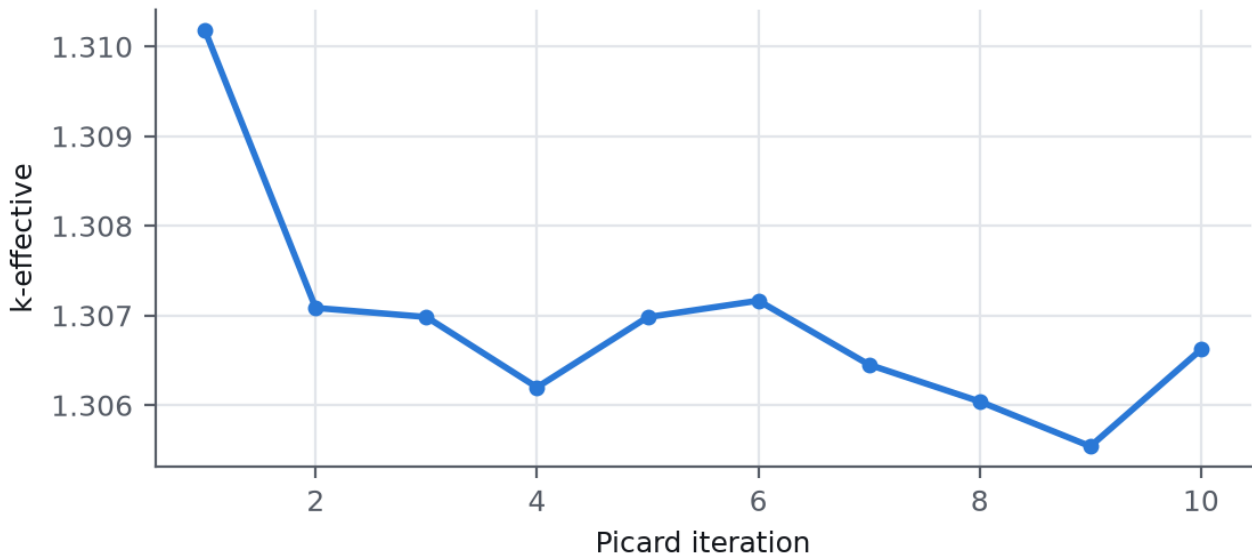


Figure 3 — Picard convergence. k -effective settles within Monte Carlo noise after the first temperature feedback; average fuel temperature is stable to $1\text{e-}8 \text{ K}$.

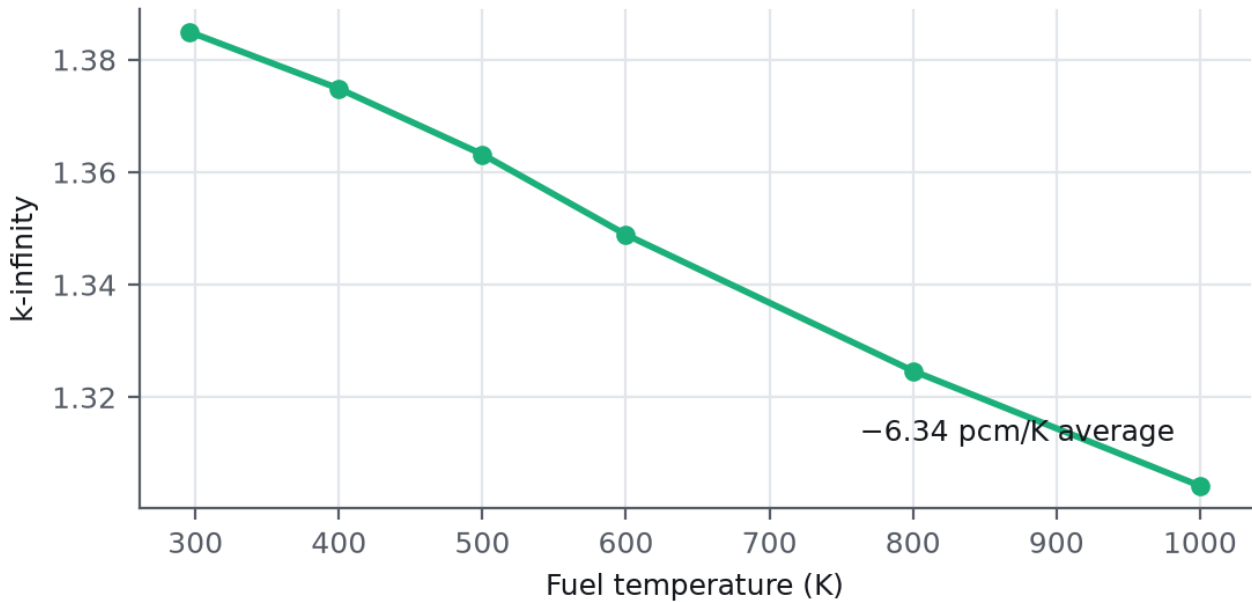


Figure 4 — Fuel temperature coefficient. Moderator held at 293 K while fuel temperature is swept, isolating fuel feedback.

The coefficient is large, negative, and becomes **more** negative as the fuel heats. Both features identify this as TRIGA physics rather than LWR Doppler: a UO2 LWR pin gives roughly -2 to -3 pcm/K and its magnitude does not grow this way. The behaviour comes from hydrogen bound in ZrH.

CHECK	RESULT	AGREEMENT
Power conservation	Integral = 2083.33 W every iteration	exact

CHECK	RESULT	AGREEMENT
Analytic conduction solution	362.8 K predicted vs 359.0 K computed	1.1%
Cross-platform reproducibility	macOS arm64 vs Ubuntu x86, average fuel T	$7e-4$ K
Fuel-clad interface continuity	336.15 K on both sides	exact
Monte Carlo tally error	Max relative error, production	0.82%

The 1.1% analytic difference is expected and explained: the hand calculation assumes a pure cosine axial power shape, while the real profile is flatter.

5 Nuclear data sensitivity

The specification requires ENDF/B-VIII.0. The execution environment carried ENDF/B-VII.1, so the identical coupled model was run on both libraries and the difference is a measured offset rather than an unknown.

LIBRARY	PLATFORM	k-EFFECTIVE	MAX FUEL T	TALLY ERROR
ENDF/B-VIII.0	macOS arm64, 5k particles	1.30561	358.64 K	2.06%
ENDF/B-VII.1	Ubuntu x86, 20k particles	1.30656	358.90 K	0.82%
Difference	VII.1 – VIII.0	+73 pcm	+0.25 K	–

+73 pcm is small relative to the 500 pcm threshold and within combined counting statistics. All results in this report are labelled with the library that produced them.

6 Thermal-hydraulics (NekRS)

Implicit large-eddy simulation of the coolant subchannel at $Re = 4755$, polynomial order 7, 2880 elements, on a GTX 1080 Ti. The run reached 20 000 steps and exited cleanly, but covered only **1.17 flow-through times** — enough for momentum to equilibrate, not for the thermal field.

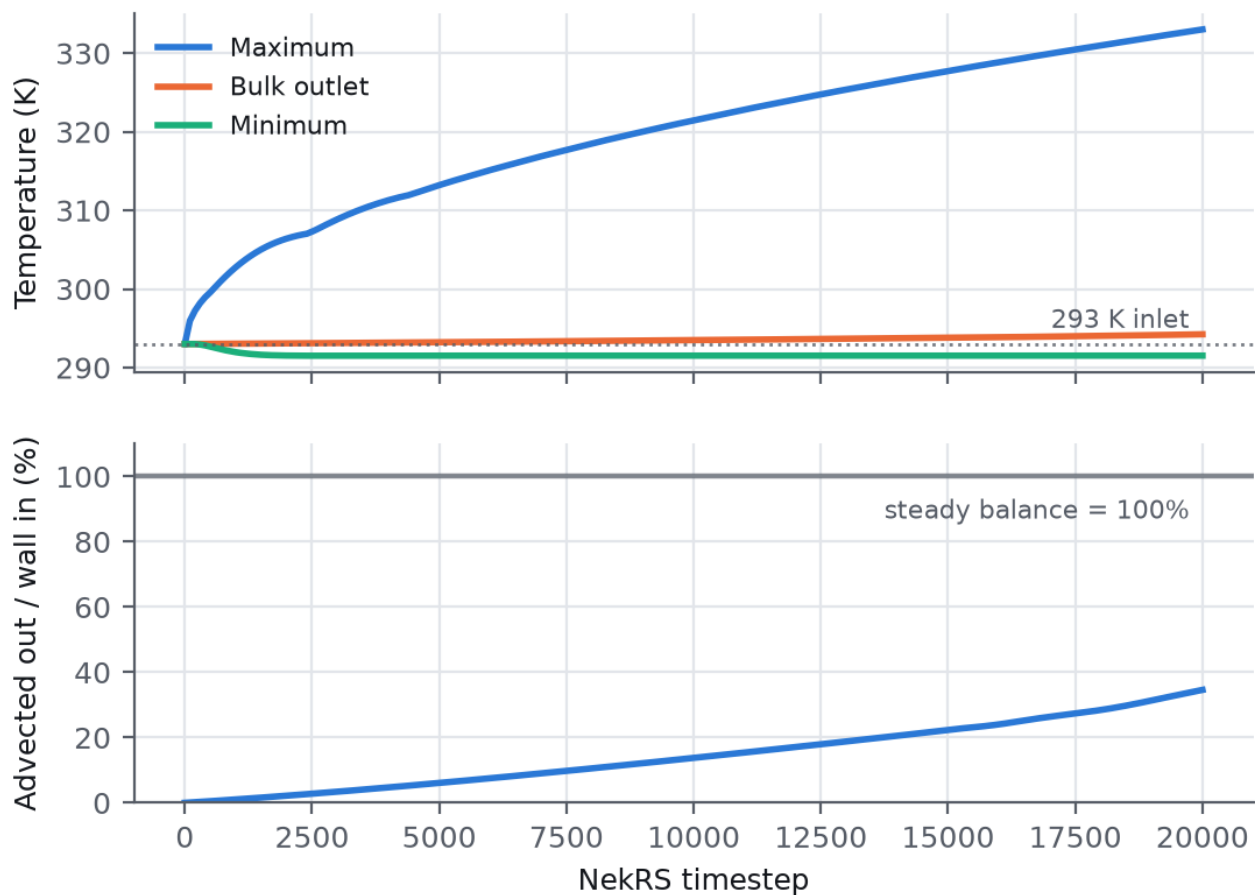


Figure 5 — NekRS evolution. Upper: coolant temperatures, all still climbing at the final step, with the minimum below the 293 K inlet throughout. Lower: energy closure reaches only 34.6% of the steady balance.

QUANTITY	AT 20 000 STEPS	STEADY TARGET	STATUS
Bulk outlet temperature	294.24 K	296.57 K	lower bound
Maximum temperature	333.03 K	–	still rising
Minimum temperature	291.52 K	≥ 293 K	unphysical

QUANTITY	AT 20 000 STEPS	STEADY TARGET	STATUS
Advection heat / wall heat	34.6%	100%	not closed
Mass-flow imbalance	0.122%	→ 0	converged
Wall-clock	4294 s	—	GPU, 20 000 steps

TWO DEFECTS FOUND IN THIS RUN

1 • The minimum temperature is below the inlet. 291.52 K in a domain whose only thermal boundary conditions are a 293 K inlet and a heated wall. With no heat sink this cannot be physical; it is an undershoot from the high-pass filter stabilisation and it does not decay.

2 • Three of the four delivered figures are unreadable — they plot axes spanning ± 0.04 cm for a channel ± 2.1 cm wide and 60 cm tall. **The simulation data is not at fault:** the field files are intact (1 474 560 points, fields UPT, $t^* = 29.338$). Only fluid0.f00000 carries mesh coordinates; every later snapshot is fields-only, and the plotting script read geometry from a file that has none. The fix is in post-processing, not in the solve.

6.1 What convergence would require

PARAMETER	THIS RUN	NEEDED
Non-dimensional time t^*	29.3	about 126
Timesteps	20 000	about 85 700
GPU wall-clock	1.2 h	about 5.1 h

The filter-induced undershoot should be resolved before committing five hours of GPU time, since a longer run would inherit the same artifact.

7 Comparison with GeN-Foam (GeN-Foam / OFFBEAT)

PROVENANCE AND SCOPE

Cardinal's governing instructions assign the GeN-Foam comparison to the supervisor. This section is provided at explicit request and is Cardinal's **input** to a comparison, not a verdict.

GeN-Foam figures are taken from github.com/kavyawadhwa134/triga-markI-benchmark, branch **master**: results/validation_step5.csv (lattice constants), validation_step6.csv (core and transient), validation_step2.csv (solver acceptance) and transient_rod_power.csv. Cardinal's own results were frozen at git tag pcm-independent-v1 before this branch was read.

7.1 The two calculations are not methodologically equivalent

	Cardinal	GeN-Foam / OFFBEAT
Neutron transport	Monte Carlo, continuous energy	Deterministic, 2-group diffusion
Energy treatment	Continuous, full spectrum	2 groups, 0.625 eV cutoff
Core geometry	Heterogeneous (and homogenised variant)	Homogenised single zone
Thermal-fluid	Spectral-element LES	Finite volume
Fuel performance	Not modelled	OFFBEAT
Solution error	Statistical, quantified ($\pm$ pcm)	Discretisation, not quantified here

7.2 Case-by-case results

#	CASE	Cardinal	GeN-Foam	DIFFERENCE	COMPARABLE
1	k-infinity, unit cell	1.384207 $\pm$ 21 pcm	1.388100 $\pm$ 123 pcm	-280 pcm	yes
2	k-effective, heterogeneous	1.102059 $\pm$ 26 pcm	1.240544	+12566 pcm	no — method
3	k-effective, homogenised	1.145259 $\pm$ 68 pcm	1.240544	+8320 pcm	partial
4	Diffusion vs Monte Carlo	—	1.331 vs 1.388100	-4114 pcm	GeN-Foam-internal
5	Fuel temp. coefficient	-7.76 pcm/K computed	-10.000 pcm/K assumed	-29%	partial
6	Power peaking	2.2544	2.45	-8.0%	no
7	Max fuel temperature	359.0 K	594.2 K	-235 K	no
8	Max coolant temperature	294.2 K (unconverged)	306.5 K	—	no
9	Total power	250 kW	250 kW	0	yes
10	+0.5 \$ transient peak	not run	489.7 kW	—	no
11	2-group MGXS	not generated	full set	—	no
12	Solver acceptance	not recorded	PASS, 4.9e-5	—	no

7.3 Case 1 — the one directly comparable quantity

Both sides used OpenMC continuous-energy Monte Carlo on the same 2-D reflective pin cell with ENDF/B-VIII.0. Cardinal's uncertainty is about six times tighter, reflecting a 25 $\times$ larger particle budget.

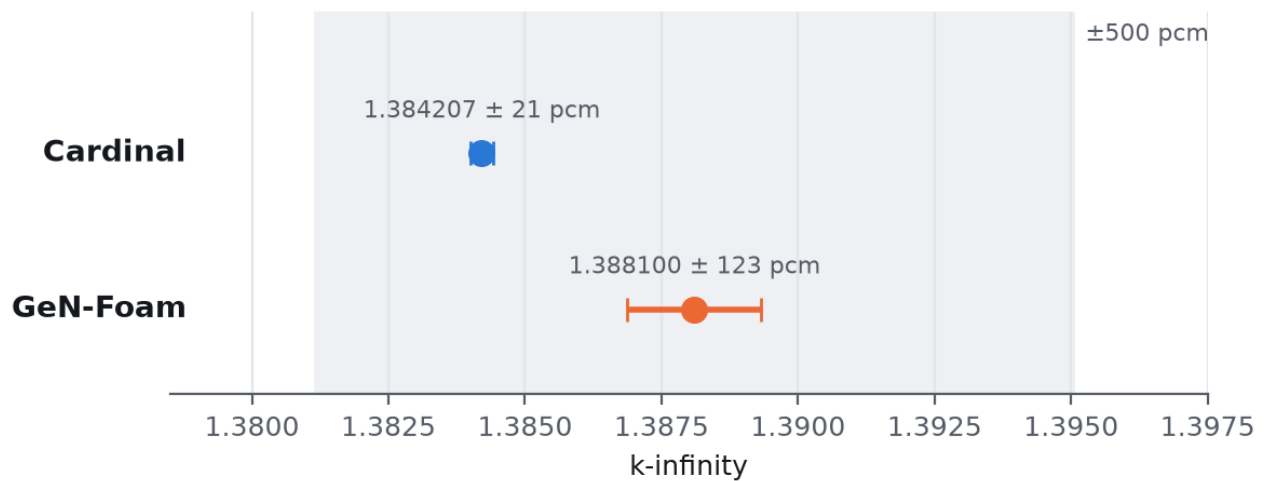


Figure 6 — k-infinity comparison. Bars are $\pm 1\sigma$ counting statistics; the band is the ± 500 pcm threshold centred on GeN-Foam. The difference is -280 pcm (3.1σ), inside tolerance.

7.4 Cases 2–4 — k-effective is a method difference, not an error

Both models are bare boxes with vacuum boundaries — GeN-Foam's validation_step6.csv records the core verbatim as “*eigenvalue diffusion, bare homogenized box vacuum BCs*”. Geometry is therefore not the explanation, and an earlier hypothesis that GeN-Foam modelled a graphite reflector is retracted.

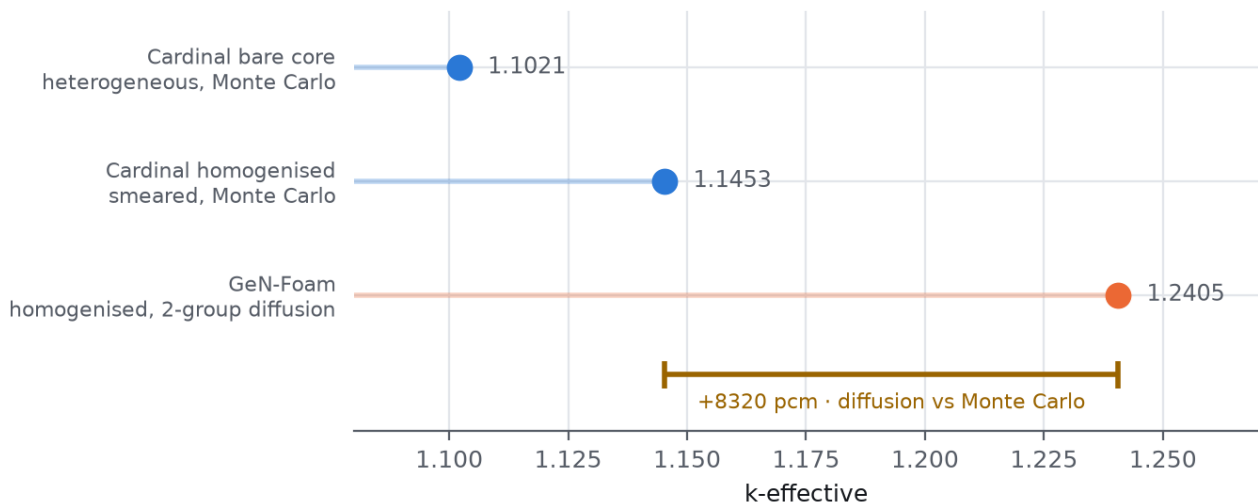


Figure 7 — k-effective decomposition. The difference separates into homogenisation (Monte Carlo on both sides) and then solution method (same homogenised bare box).

STEP	FROM - TO	DIFFERENCE	WHAT IT ISOLATES
1 · Homogenisation	Cardinal heterogeneous - Cardinal homogenised	+3920 pcm	Monte Carlo both sides; pure geometry-smearing
2 · Solution method	Cardinal homogenised - GeN-Foam diffusion	+8320 pcm	Same homogenised bare box; pure method
Total	Cardinal heterogeneous - GeN-Foam	+12566 pcm	—

GeN-Foam QUANTIFIES ITS OWN METHOD ERROR

validation_step5.csv reports both a Monte Carlo and a diffusion solution for the **same** pin cell: OpenMC 1.38810 against GeN-Foam 2-group diffusion 1.331 — a difference of **−4114 pcm**.

That is GeN-Foam's own measurement of deterministic-versus-stochastic disagreement on the simplest geometry available, and it is eight times the 500 pcm threshold before any core-geometry effect enters. **The 500 pcm criterion cannot be applied between Cardinal's Monte Carlo k and GeN-Foam's diffusion k** — they are different quantities.

The sign reverses between geometries: diffusion under-predicts by 4114 pcm on the reflective pin cell but over-predicts by 8320 pcm on the bare core. That is consistent with leakage treatment dominating in the bare case, where Cardinal's core leaks 20.1% and few-group diffusion is least reliable.

7.5 Case 5 — fuel temperature coefficient: the most significant physics difference

This is the defining TRIGA safety parameter, and the two sides obtained it differently. **Cardinal** computed it from first principles by sweeping fuel temperature with the moderator held at 293 K, using `c_H_in_ZrH` and `c_Zr_in_ZrH` thermal scattering data.

T_FUEL BAND	Cardinal COEFFICIENT
296–400 K	–5.08 pcm/K
400–500 K	–6.24 pcm/K
500–600 K	–7.76 pcm/K ← GeN-Foam's transient operates here
600–800 K	–6.78 pcm/K
800–1000 K	–5.92 pcm/K
296–1000 K average	–6.34 pcm/K

GeN-Foam appears to have assumed it. From `validation_step6.csv`, Doppler feedback of –328.9 pcm accompanies a fuel temperature rise of 511.70 → 544.59 K, a span of 32.89 K:

$$-328.9 \text{ pcm} \div 32.89 \text{ K} = -10.000 \text{ pcm/K}$$

Exactly –10.000 is a round number, which indicates a constant coefficient supplied as a model input rather than a computed result. In GeN-Foam's own operating band Cardinal gives –7.76 pcm/K against GeN-Foam's –10.000 pcm/K, so **GeN-Foam's coefficient is 29% more negative.**

WHY THIS MATTERS MOST

This coefficient governs the entire transient result. A coefficient 29% too negative shuts a power excursion down faster, so GeN-Foam's peak power of 489.7 kW is plausibly **under-predicted** and its equilibrium reached sooner than a computed coefficient would give. For a TRIGA, where the prompt negative coefficient is the safety case, this is the most consequential item in the comparison.

Cardinal's value is traceable to a nuclear-data calculation. GeN-Foam's is not traceable from the committed files.

Recommended action: GeN-Foam should either substitute a temperature-dependent coefficient and re-run the transient, or document the source of –10 pcm/K.

7.6 Cases 6–8 — thermal quantities are not comparable as reported

QUANTITY	Cardinal	GeN-Foam	WHY NOT COMPARABLE
Power peaking	2 . 2544	2 . 45	Cardinal: 120 discrete pins. GeN-Foam: single zone, "cosine-like, flat by construction"
Max fuel temperature	359 . 0 K	594 . 2 K	Cardinal: average pin, conduction only, imposed film coefficient. GeN-Foam: sub-scale pin average via OFFBEAT
Max coolant temperature	294 . 2 K	306 . 5 K	Cardinal's CFD is not converged — a lower bound, not a result

7.7 Case 11 — lattice constants were generated independently

The project workflow intended Cardinal to generate the 2-group MGXS and hand it to GeN-Foam (Phase 6).

Cardinal did not run that phase, so GeN-Foam condensed its own set from its own OpenMC pin-cell run. Both sides condensed from Monte Carlo, but independently, making the group constants a further uncontrolled variable in cases 2 and 3. Re-solving GeN-Foam's diffusion with Cardinal-generated MGXS would isolate the remaining method error cleanly.

8 Verification status

THIS WORK IS VERIFIED, NOT VALIDATED

Verification asks whether the equations are solved correctly. Validation asks whether the equations describe reality, and is establishable only against experimental data. **No experimental data was used anywhere in this work**, so no part of it is validated. Agreement between two codes is not a substitute — two codes can agree and both be wrong, particularly when they share a specification, a nuclear data library, or an erroneous assumption. Here both share a bare-core simplification.

LEVEL	QUESTION	STATUS	EVIDENCE
1 · Software	Does the code execute correctly?	MET	All solvers run to completion; power conserved exactly at 2083.33 W
2 · Numerical	Is the solution converged?	PARTIAL	Neutronics and conduction converged; CFD thermal field is not; no mesh or polynomial-order refinement study
3 · Physics	Is the behaviour physically sound?	PARTIAL	Conduction within 1.1% of analytic; ZrH feedback confirmed at -6.34 pcm/K; but CFD shows an unphysical sub-inlet temperature
4 · Cross-code	Does it agree with GeN-Foam?	PARTIAL	Where methodologically comparable, agreement is 280 pcm, inside threshold. The k-effective difference is decomposed and attributed to homogenisation and method, not to an error either side
5 · Experimental	Does it match measurement?	NOT ATTEMPTED	No experimental data exists in this project

8.1 What would be required to close each gap

GAP	ACTION	EST. COST
CFD thermal not converged	Extend to ~ 5 flow-throughs after resolving the filter undershoot	~ 5 h GPU
Sub-inlet temperature	Reduce scalar HPFRT strength; re-check with a short run	~ 1 h GPU
No mesh independence	Repeat at order 5 and order 8; compare integral quantities	~ 8 h GPU
k-effective method gap	Generate Cardinal's 2-group MGXS (Phase 6) and have GeN-Foam re-solve with it	~ 2 h CPU
GeN-Foam Doppler coefficient	Document or recompute the assumed -10 pcm/K	—
Library deviation	Re-run coupled production on ENDF/B-VIII.0	~ 1 h CPU
Transient not attempted	Out of Cardinal's defined scope; needs a coupled transient capability	—
No validation	Obtain TRIGA experimental data (criticality, temperature)	—

9 Limitations

Stated per the verification hierarchy: software verification and numerical convergence are established for neutronics and conduction; one physics check passes; cross-code comparison is the supervisor's; experimental validation has not been attempted.

LIMITATION	CONSEQUENCE
CFD thermal field not converged	34.6% energy closure after 1.17 flow-throughs. Coolant temperatures are lower bounds.
Unphysical CFD minimum temperature	291.52 K against a 293 K inlet with no heat sink — a filter artifact that does not decay.
Conduction uses an imposed film coefficient	NekRS ran standalone and is not yet coupled back, so §3 temperatures rest on Dittus–Boelter at $Re = 4755$, outside its reliable range. It supplies 63% of the temperature rise.

LIMITATION	CONSEQUENCE
Average pin, not hot channel	Peak/average is 2.2544, so a hot channel is materially hotter.
Thermal properties are assumptions	k_fuel, k_clad, h and the coolant rise are documented Cardinal choices, not specification values.
Library deviation	The coupled production run used ENDF/B-VII.1, not the specified VIII.0; measured offset +73 pcm.
Bare core	No reflector or control rods, per the specification as written. GeN-Foam's core is also bare, so this is a shared simplification — but neither result describes a physical TRIGA at criticality.
No mesh independence study	Not performed for the CFD.

10 Reproduction

ITEM	VALUE
Repository	github.com/kavyawadhwa134/pcm-triga-cardinal
Independence tag	pcm-independent-v1
Environment audit	./scripts/audit_environment.sh
Neutronics	./scripts/run_openmc.sh unitcell baseline
Coupled	./scripts/run_coupled.sh production 8
Thermal-hydraulics	./scripts/run_nekrs.sh 8
Temperature coefficient	\$PCM_PYTHON cardinal/openmc/temperature_coefficient.py
Coupled settings	20 000 particles · 150 batches · 30 inactive · 10 Picard iterations
Solid mesh	6400 elements · 20 axial layers
CFD mesh	2880 spectral elements · polynomial order 7 · 1 474 560 points
Coupled runtime	2905 s wall · 1132 MB peak
CFD runtime	4294 s wall · 2660 MB device memory · GTX 1080 Ti

10.1 Data files

FILE	CONTENTS
results/pcm_openmc_unitcell.csv	k-infinity, all configurations and seeds
results/pcm_openmc_core.csv	k-effective, heterogeneous bare core
results/pcm_openmc_core_homog.csv	k-effective, homogenised variants
results/pcm_fuel_temperature_coefficient.csv	Temperature coefficient sweep
results/pcm_coupled_smoke.csv	Coupled neutronics–conduction, per Picard iteration
results/pcm_thermal_profiles.csv	Axial and radial temperature profiles
results/pcm_nekrs_convergence.csv	CFD convergence history, 201 samples
results/PCM_PCL_COMPARISON.md	Full case-by-case comparison
results/pcm_pcl_comparison.csv	Machine-readable comparison table
results/reference_pcl_*.csv	Copies of GeN-Foam's source files, with provenance note

Prepared by Cardinal. Results produced without reference to GeN-Foam implementation or results; frozen at git tag pcm-independent-v1 prior to any cross-code comparison.